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Requirements Engineering Analysis
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Riyadh Metro Ticket Booking

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Introduction

Riyadh Metro provides efficient solutions for people to travel in comfortable metro cars equipped with the latest international technologies. Riyadh metro has many advantages for example, when a person travels from home to university or workplace, he\she won't facing a daily traffic or the problem of finding a parking for your car. Our application will focus on booking tickets system, the users can sign in to the system easily, all they have to do is fill in the information and the system will verify the validity of it. The users must be registered in the application in order to be able to book the ticket, they can get more than the one ticket and book it at any time, all they have to do to book is entering the details and selecting the landing station, also they can cancel or modify their ticket, after that the user enter the details of the payment process and a verification message which is sent to the user, and your ticket will be available.

System scope

The system will be used for booking Riyadh metro tickets and it's scope will be within the Riyadh city only, 1.6 million tickets is the maximum number of tickets in sell per day, while the maximum number of users of the application will be 5000 users. This application will be launched in late 2022.

Stakeholders

Administrator (Royal Commission for Riyadh City) will have control of all the system

User: The person who uses the application to book tickets.

Staff including:

Customer Service: For helping the users in solving the problems they face, such as solving reservation problems.

Technicians: The ones who monitor the progress of the application and take care of its development and can also solve any technical problems facing the application

Interviews

We conducted interviews with several stakeholders interviewed at the same time to save time and efforts. The interview's questions will be in the last page.

Survey

This survey was used to achieve the largest number of suggestions from people who have tried the ticket booking application, to use it in writing a good requirements to develop an excellent application!

This survey was created on 02/10/2021, during these weeks, 60 people had fill in the survey, and it was published in social media to get most responses. Most of them concluded that they did not face any difficulties or problems in booking, on the other hand, some of them get their booking canceled due to an electronic error, they provided a good suggestion that can be used to develop the application.

Https://forms.gle/55qkmhtaz2cyfxd76

JAD

We choose this technique because it facilitated the JAD session that bring Participants together to produce high-quality deliverables in a short time period.

The sessions were held for four hours per day for two weeks, the sponsor who makes the final decisions, determines the leader of the session, which will be run by the leader The sponsor and leader first task is to determine the goal of the session which gather the stakeholders in one place to determine the functional and non-functional requirements and the brainstorming technique was used to collect them, the participants in the session are sponsor, leader, user, writer and staff.

Functional requirements

1. The administrator shall be able to set new tickets

1.1 the system shall be able to set the ticket number.

1.2 The system shall be able to set the ticket price for adults and children.

1.3 The system shall be able to give the administrator a list of tickets sorted by date.

2. The user shall be able to create an account.

2.1 The user shall be able fill his/her personal information.

2.2 The system shall send a phone message to the user to verify the personal information.

3. The user shall be able to log in.

3.1 The system shall allow the user to request a new password in case of forgetting it.

3.1.1 The system shall send a verification message to the e-mail.

3.1.2 The user shall enter the new password.

4. The user shall be able to search for the previous ticket.

4.1 The system shall show the user's previous ticket that sorted by date.

5. The system shall allow the user to modify the day's ticket.

5.1 The system shall allow the user to modify the ticket details.

5.2 The system shall allow the user to modify the stations.

6. The user shall be able to select the ticket information.

6.1 The system shall allow the user to search for the date of the ticket.

6.2 The user shall select the main arrival station.

6.3 The system shall display the user a list of substations.

6.4 The user shall determine his\her personal location.

6.5 The system shall display all the closet stations to the user.

7. The user shall be able to book the ticket.

7.1 The system shall be able to allow the user to select more than one ticket.

7.2 The system shall allow the user to select the age group.

8. The user shall pay the ticket price.

8.1 The user shall enter the payment information.

8.2 The user shall enter the confirmation code for the payment process.

9.The system shall display the booking information after payment.

9.1 The system shall send a message to the user's e-mail about his\her booking information.

10.The system shall allow the user to be able to communicate with the customer service employee.

10.1 The system shall display the user's communication box .

10.2 The system shall allow the employee to display the user's booking information.

Non-functional requirements

Performance :

It should not be possible to double-book tickets (i.e., buy a ticket that has already -been sold).

- Responses to inquiries should not take more than 7 seconds to load on the screen after the user submits the query.

- The transition between pages should be in less than 3 seconds.

- Error messages will be displayed if the user entered an invalid value.

- No more than 10 numbers must be entered in the mobile number field.

- The system shall send the verification message to the user's number for a maximum of 30 seconds.

Safety:

- The system shall return the user to the booking page automatically if the inactivity period exceeds 5 minutes on the payment page

Reliability:

- All data is backed up daily.

Privacy:

- Only registered customers can book tickets.

Availability:

- Users shall be able to book tickets 24/7

- The system shall display the availability of tickets within five seconds.

Portability:

- The system supports IOS and Android operating systems.

Capacity;

- the capacity of the application users is a maximum of 5,000 users per day.

- The capacity for the number of tickets does not exceed 10 tickets.

Interface:

- The system shall provide on the registration page user information fields .

- The system should provide a confirmation button on the registration page that will direct the user to the ticket purchase page.

- The system should provide the ticket book details fields on the ticket booking page.

- The system should provide a search box on the My Tickets page.

- Previous ticket information should be in black.

- Current ticket information should be in gray.

- The system should provide a confirmation button at the end of the ticket page that will direct the user to the payment page.

- The credit card details fields should appear on the payment page

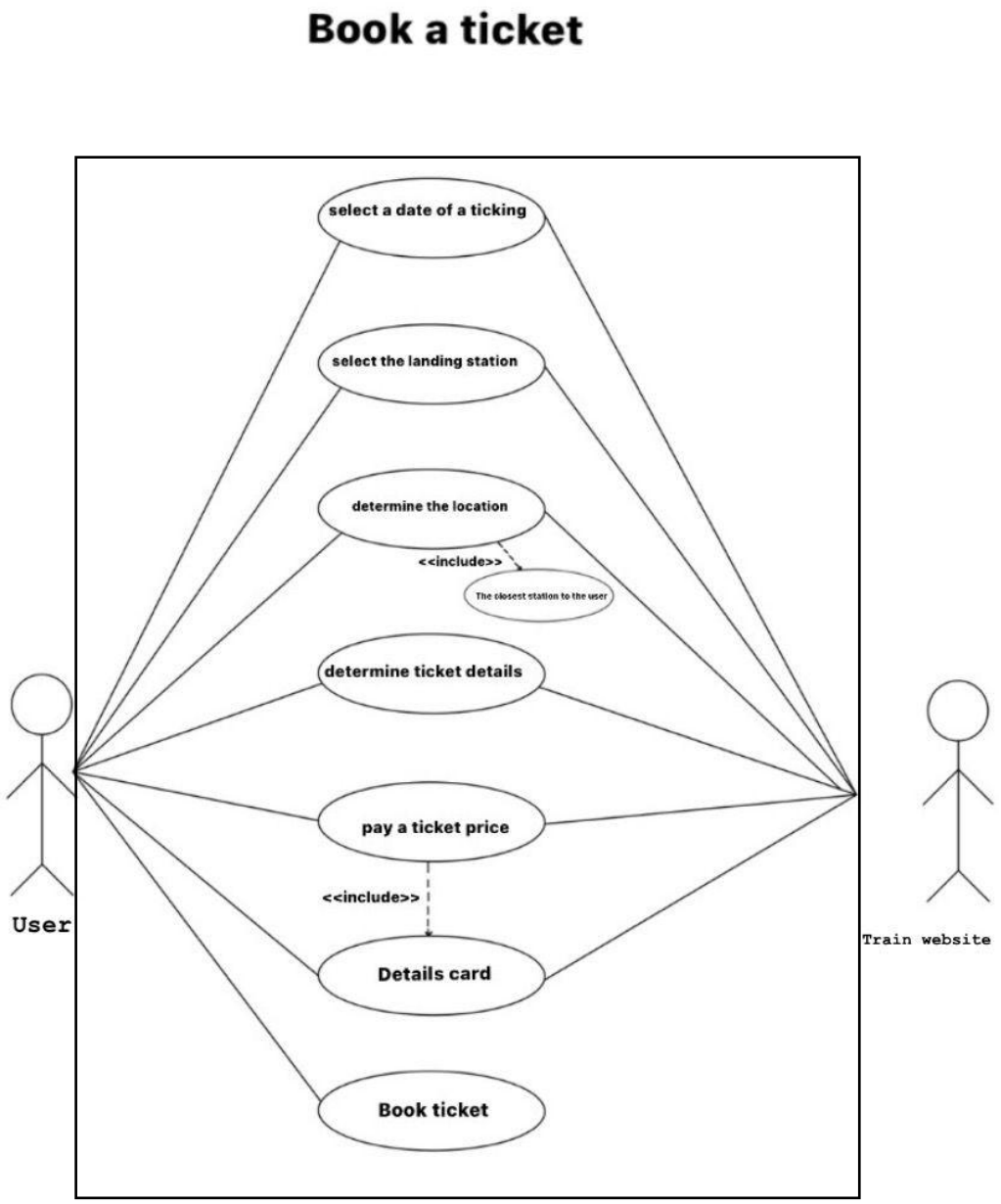
(Type, number, CVV number, expiration date and name on the card) .

User interfaces

Https://www.canva.com/design/daexahstwho/9zvaiwifxh_l4f5hgthpua/view?Mode=prototype

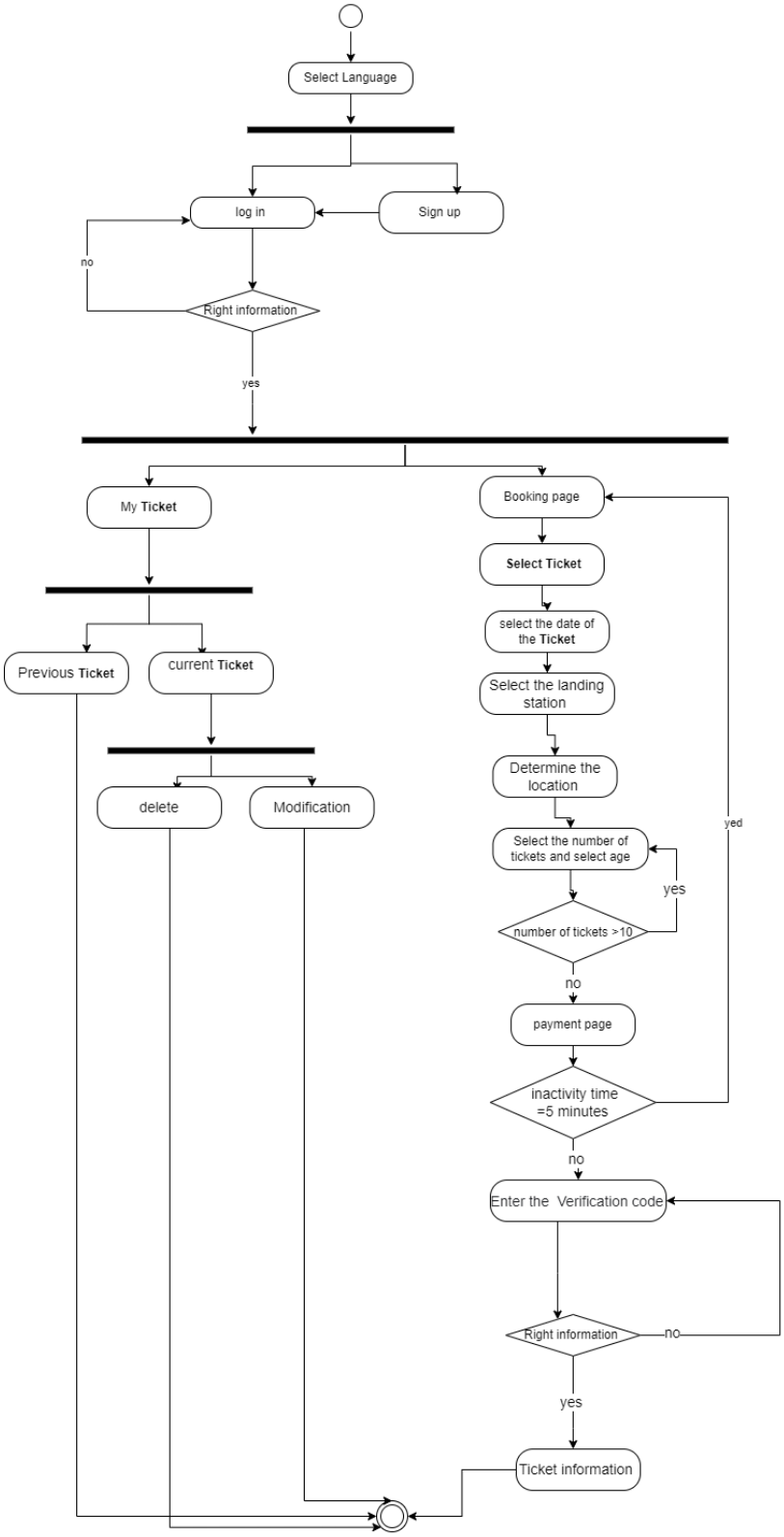
Use case diagram

We used a use case diagram to describe the scenarios (or sequence of actions) in the process of booking a ticket in the system and to show how the actor interacts with the system.



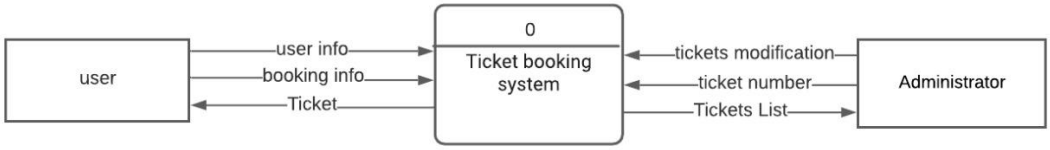
Activity diagram

We used the activity diagram to illustrate the flow of control from one activity to another for the entire system.



Context diagram

One process that represents the system which is the booking system and how it interacts with external entities and they are the user and the Administrator



V&V

Test-case generation technique:

Test case on requirement TK-1: " No more than 10 numbers must be entered in the mobile number field".

Scenario	Test Step	Expected Result	Actual Outcome
Verify that the "phone number" field that can accept maximum of 10 numbers	Writing a numbers in field "phone number" in 10 numbers	The Application should be able to accept all 10 numbers.	The Application accepts all 10 numbers.
Verify that the "phone number" field that can accept maximum of 11 numbers	Writing a numbers in field "phone number" in 11 numbers	Application should NOT accept all 11 numbers.	The Application accepts all 10 numbers.

Test case on requirement TK-2:"The capacity for the number of tickets does not exceed 10 tickets".

Scenario	Test Step	Expected Result	Actual Outcome
Verify that if the ticket selection process can accept a maximum of 10 tickets	Select 10 tickets	Application should be able to accept all 10 tickets.	Application accepts all 10 tickets.
Verify that if the ticket selection process can accept a maximum of 11 tickets	Select 11 tickets	Application should NOT accept all 11 tickets.	Application accepts all 10 tickets.

Test case on requirement TK-3: " The system shall return the user to the booking page automatically if the inactivity period exceeds 5 minutes on the payment page".

Scenario	Test Step	Expected Result	Actual Outcome
Verify that the time in inactivity process cannot exceed a maximum of 5 minutes in the payment page	The time in inactivity process do not exceed a maximum of 5 minutes in the payment page	Application should be able to accept that the time in inactivity process do not exceed a maximum of 5 minutes in the payment page	Application accepts that the time in inactivity process do not exceed a maximum of 5 minutes in the payment page
Verify that the time in inactivity process can exceeds a maximum of 5 minutes in the payment page	The time in inactivity process exceed a maximum of 5 minutes in the payment page	Application should NOT accept that the time in inactivity process exceed a maximum of 5 minutes in the payment page	Application accepts that the time in inactivity process do not exceed a maximum of 5 minutes in the payment page

Test case on requirement TK-4 " Verify that the capacity of the application users is a maximum of 5,000 users per day".

Scenario	Test Step	Expected Result	Actual Outcome
Verify that the capacity of the application users can be maximum of 5000 users per day	5000 users per day Using the application per day	Application should be able to accept all 5000 users.	Application accepts all 5000.
Verify that the capacity of the application users can be maximum of 5500 users per day	5500 users per day Using the application per day	Application should NOT accept all 5500 users.	Application accepts all 5000.

Test case on requirement TK-5" The system must send a verification message to the user's number in a maximum of 30 seconds".

Scenario	Test Step	Expected Result	Actual Outcome
Verify that the system can send a verification message to the user's number within 30 seconds as a maximum.	Send a verification message to the user's number within 30 seconds.	Application should be able to send a verification message to the user's number within 30 seconds	Application accepts to send a verification message to the user's number within 30 seconds
Verify that the system can send a verification message to the user's number within 1 minute as a maximum.	Send a verification message to the user's number within 1 minute.	Application should NOT accept that the verification message is sent to the user's number within 1 minute	Application accepts to send a verification message to the user's number within 30 seconds

Test case on requirement TK-6 "It should not be possible to double-book tickets (i.e. Buy a ticket that has already been sold)".

Scenario	Test Step	Expected Result	Actual Outcome
Verify that the booking of a ticket has not been booked before is accepted	Buy a ticket that is has not been booked before	Application should be able to accept the booking of a ticket that has not been booked before	Application accepts the booking of a ticket that has not been booked before
Verify that the booking of a ticket has been previously booked is accepted	Buy a ticket that is booked before	Application should NOT accept the booking of a ticket that has been booked before	Application accepts the booking of a ticket that has not been booked before

Test case on requirement TK-7 "The transition between pages should be in less than 3 seconds".

Scenario	Test Step	Expected Result	Actual Outcome
Verify if transition between pages can be less than 3 seconds	Transition between pages takes less than 3 seconds	Application should be able to accept that the transition between pages takes less than 3 seconds	Application accepts that the transition between pages takes less than 3 seconds
Verify if transition between pages can be more than 3 seconds	Transition between pages takes more than 3 seconds	Application should NOT be able to accept that the transition between pages takes more than 3 seconds	Application accepts that the transition between pages takes less than 3 seconds

Test case on requirement TK-8" Responses to inquiries should take not more than 7 seconds to load on the screen after the user submits the query ".

Scenario	Test Step	Expected Result	Actual Outcome
Verify if responses to queries cannot take more than 7 seconds to load on the screen after the user submits the query	Responses to queries do not take more than 7 seconds to load on the screen after the user submits the query	Application should be able to accept that responses to queries do not take more than 7 seconds to load on the screen after the user submits the query	Application accepts that responses to queries do not take more than 7 seconds to load on the screen after the user submits the query
Verify if responses to queries can takes more than 7 seconds to load on the screen after the user submits the query	Responses to queries takes more than 7 seconds to load on the screen after the user submits the query	Application should NOT accept that responses to queries takes more than 7 seconds to load on the screen after the user submits the query	Application accepts that responses to queries do not take more than 7 seconds to load on the screen after the user submits the query

Test case on requirement TK-9" Only registered users can book tickets".

Scenario	Test Step	Expected Result	Actual Outcome
Verify that only registered users can book tickets	Registered users book a ticket	Application should be able to accept that registered users can book tickets	Application accepts that registered users can book tickets
Verify that unregistered users can book tickets	Unregistered users book a ticket	Application should NOT be able to accept that unregistered users can book tickets	Application accepts that registered users can book tickets

A Manual Test Case Technique:

Test case on requirement TK-10" The user shall be able to log in".

Id	Summary	Steps to reproduce:	Expected Result	Actual Outcome	Status
TK-10	Check log in with valid info	1-Open the app 2-Enter the info 3- Press on the next button	The user is logged in the app	The user is logged in the app	Success
	Check log in with invalid info	1-Open the app 2-Enter the info 3- Press on the next button	The warning message is displayed	The warning message is displayed	Success

Test case on requirement TK-11" The user shall enter the confirmation code for the payment process".

Id	Summary	Steps to reproduce:	Expected Result	Actual Outcome	Status
TK-11	Check confirmation with valid number	1-Open the app 2- Log in 3- Enter tickets details 4- Enter the payment details 5- Enter the confirmation number 6- Press the next button	The user gets the ticket	The user gets ticket	Success
	Check confirmation with invalid number	1-Open the app 2- Log in 3- Enter tickets details 4- Enter the payment details 5- Enter the confirmation number 6- Press the next button	The warning message is displayed	The warning message is displayed	Success

Summary

We developed this system to make an easy booking for people. , in analysis phase we identified 4 things and they are : scope, stakeholders, how we gathered the requirements and the functional and non-functional requirements . In the system scope, we wrote the city, as well as the maximum value of the application users, the number of booked tickets. We also identified the stakeholders who they are administrator, user and staff, we also gathered the reequipments using the JAD technique in which we used brainstorming technique, there are other techniques that we used which are the interview and the survey. We also wrote the functional and non-functional requirements, in the design phase we represented the reequipments using the activity diagram to explain the entire system, and the use case diagram to explain the booking process and context diagram to explain how the external entities interact with system, we also designed the user interface, in V&V phase we used test-case generation technique and a manual test case to test technique the requirements.

Interview questions:

- What is your vision for the booking a tickets system?

  Booking a tickets system is useful for people, it saves time and effort.

- Do you think it is difficult to book from the app?

  No, because all you need is to fill in the information that the application needs.

- What it is the information that is required to register in the application?

  Name, mobile number, National identity, e-mail and password.

- Does a person have to be registered in the application in order to book a ticket?

  Yes, he\she must.

- How will the information that entered in the application be validated?

  By sending a verification message to the entered mobile number.

- Can a person book a ticket at any time?

  Yes, if you find enough capacity.

- Can the users cancel the tickets or modify it?

  Yes, they can.

- When the user cancels the tickets, what will happen to the ticket price?

  The price will be returned to the user's wallet

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